

HOTEL MANAGEMENT PLATFORM

Overview

The Hotel Management Platform is a desktop-based application designed to streamline hotel reservation and room management processes. Built using JavaFX and PostgreSQL, the system provides customers with an easy-to-use interface for booking rooms, accessing hotel services, and managing reservations while maintaining accurate records through database integration.

Objective

To develop a reliable hotel management system that automates reservation handling, room allocation, customer management, and checkout operations. The project aims to demonstrate the practical implementation of Java programming, JavaFX GUI development, JDBC connectivity, Object-Oriented Programming principles, and database management concepts.

Features

- User Registration and Authentication
- Secure Login System
- Room Availability Management
- Multi-Room Booking
- Automatic Room Allocation
- Booking Summary Generation
- Customer Dashboard
- Room Checkout Functionality
- Food Ordering Service
- Laundry Scheduling Service
- Customer Support / Help Desk
- PostgreSQL Database Integration

Technology Stack

Programming Language: Java

Frontend: JavaFX, FXML

Database: PostgreSQL

Database Connectivity: JDBC

Build Tool: Maven

Version Control: Git

IDE: IntelliJ IDEA

System Architecture

The application follows a layered architecture consisting of:

Presentation

- JavaFX User Interface
- FXML-based screens

- User interaction handling

Logic

- Controllers
- Booking validation
- Room allocation logic
- Pricing calculations

Data Access

- JDBC Connectivity
- BookingDAO
- Database utility classes

Database

- PostgreSQL Database
- Booking records
- Room inventory
- Room allocation records

Backend Structure

The backend is implemented using **Java** and **JDBC**.

Controllers

1. HomeController

Manages application navigation and provides access to:

- Login
- Registration
- Room Booking
- Dashboard

2. SignupController

Handles customer registration and validation.

- Customer registration
- Input validation
- Booking profile creation
- Database insertion

3. LoginController

Manages user authentication.

- Credential verification
- Session creation
- Dashboard access

4. RoomSelectionController

Handles room booking operations.

- Room type selection
- Availability checking
- Stay duration selection
- Booking creation

5. SummaryController

Generates booking summaries.

- Cost calculation
- Reservation review
- Booking confirmation

6. DashboardController

Provides booking management capabilities.

- View booking details
- View room assignments
- Track reservation status

Application State Management

AppState.java

Acts as a centralized state manager.

- Maintain current user session
- Store booking information
- Share data between controllers
- Preserve navigation state

Database.java

Responsible for:

- PostgreSQL connection management
- Database initialization
- Table creation
- Room inventory setup

BookingDAO.java

Handles all database operations including:

- User registration
- Authentication
- Booking creation
- Room assignment
- Booking retrieval
- Status management

Object-Oriented Programming Concepts

The project demonstrates multiple OOP principles.

Encapsulation - Booking information is protected using private attributes and controlled access methods.

Inheritance - Room classes inherit common behaviour from abstract base classes.

Abstraction - Abstract room definitions allow specialized implementations for different room categories.

Polymorphism - Different room types implement their own pricing and room-specific logic.

Interfaces - Additional facilities such as amenities are defined using interfaces.

Database Design

The system uses **PostgreSQL** as the backend database.

1.Bookings Table

Stores customer reservation information.

Attributes

- booking_id
- name
- phone_number
- password
- date_of_birth
- address
- check_in_date
- check_out_date
- status

Status Values

- ACTIVE
- CHECKED_OUT

2.Rooms Table

Stores hotel room inventory.

Attributes

- room_no
- room_type
- is_available

Room Types

- SINGLE
- DOUBLE
- TRIPLE

Expected room distribution:

- Single Rooms: 30
- Double Rooms: 40
- Triple Rooms: 30

Total Rooms: 100

3.Booked_rooms Table

Stores room allocation information.

Attributes

- booking_id
- room_no

This table establishes the relationship between bookings and allocated rooms.

Key Functionalities

1.User Registration

Customers provide:

- Name
- Date of Birth
- Address
- Phone Number

- Password

Validation is performed before storing records.

2.Login Authentication

Users log in using:

- Booking ID
- Phone Number
- Password

Credentials are validated against database records.

3.Room Reservation

Customers can:

- Select room type
- Choose number of rooms
- Select stay duration
- Check room availability

4.Booking Summary

The system generates a detailed summary containing:

- Room details
- Check-in date
- Check-out date
- Number of days
- Booking cost
- Reservation details

5.Room Allocation

The system automatically:

- Assigns available rooms
- Updates room availability
- Creates booking-room mappings

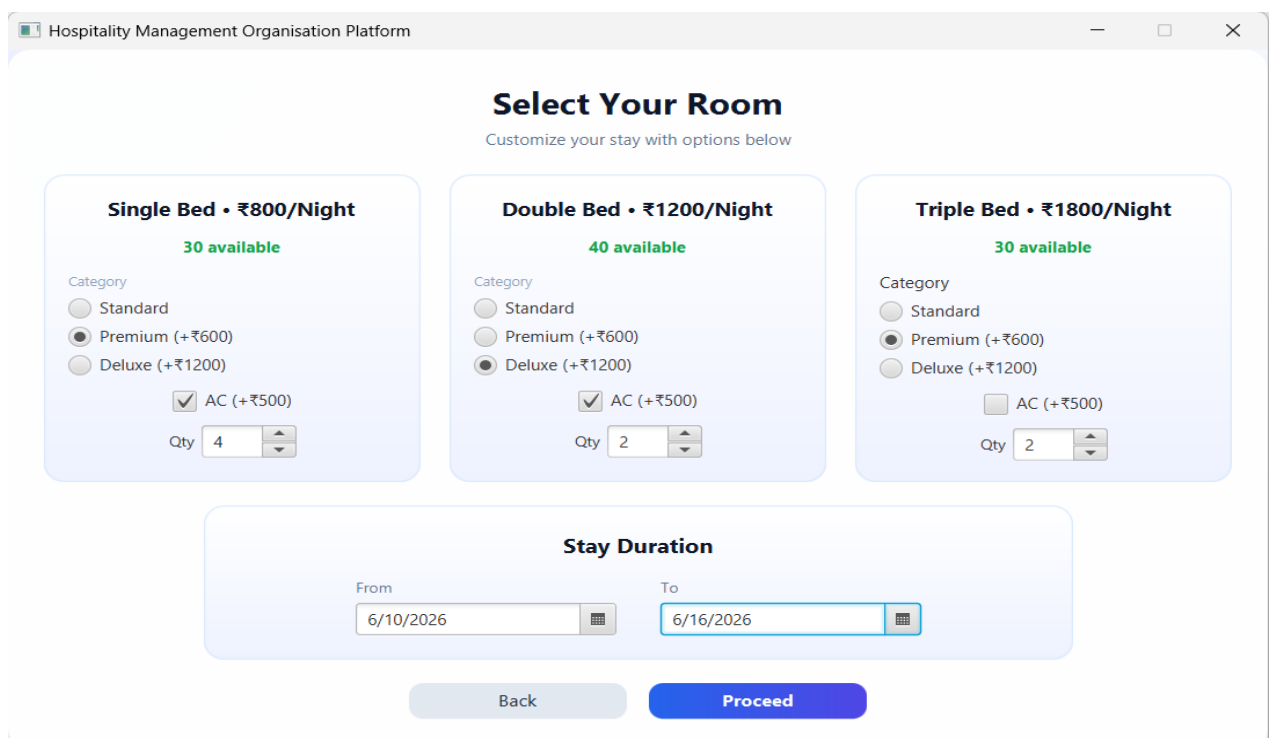
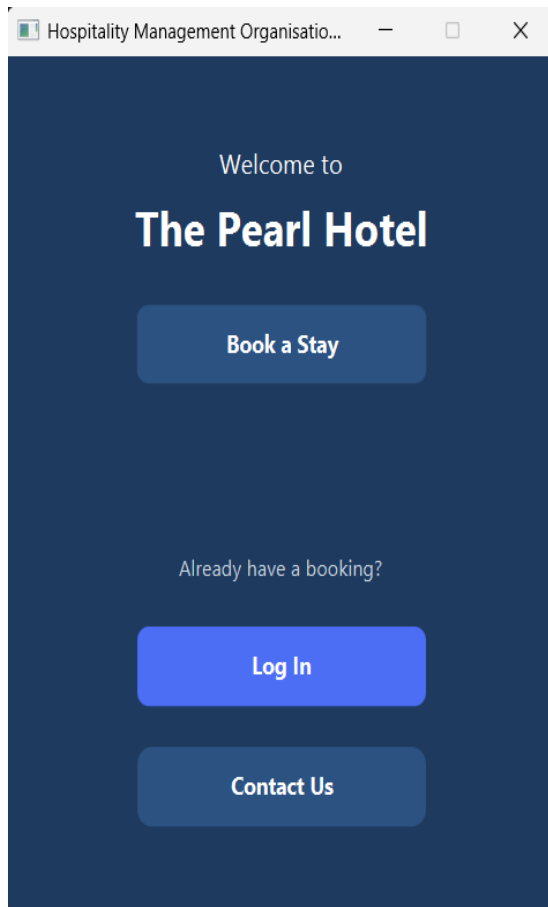
6.Dashboard

After login, customers can:

- View booking information
- Check assigned rooms
- View reservation status
- Access booking history

Frontend Structure

The application interface is developed using JavaFX and FXML.



Hospitality Management Organisation Platform

Booking Summary

Review your selections

Single Room • AC + Category • Qty: 4 ₹1900 × 4 × 6 day(s)	₹45600
Double Room • AC + Category • Qty: 2 ₹2900 × 2 × 6 day(s)	₹34800
Triple Room • AC + Category • Qty: 2 ₹2400 × 2 × 6 day(s)	₹28800

Total: ₹109200

Back

Proceed

Hospitality Management Organisati...

Guest Details

Enter your information

Full Name

Veer

Date of Birth

6/5/1980

Phone Number

9209109111

Address

#60, 4th block , JP nagar, Bengaluru-101789

Set Password

...

Confirm Password

...

Back

Book

Booking Confirmed

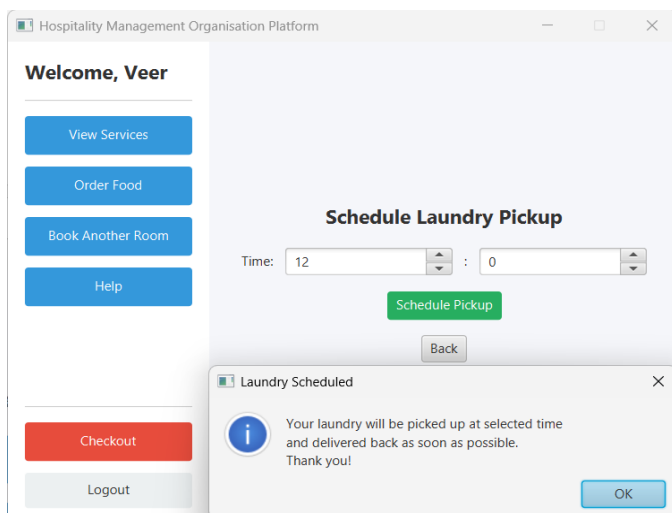
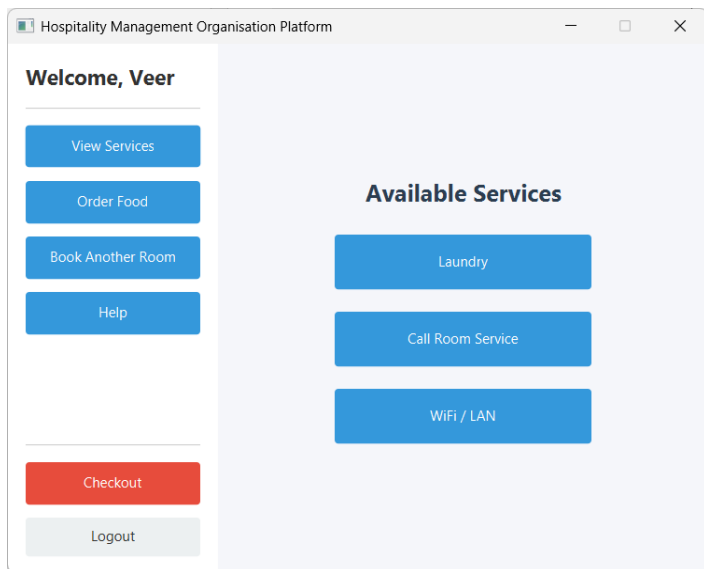
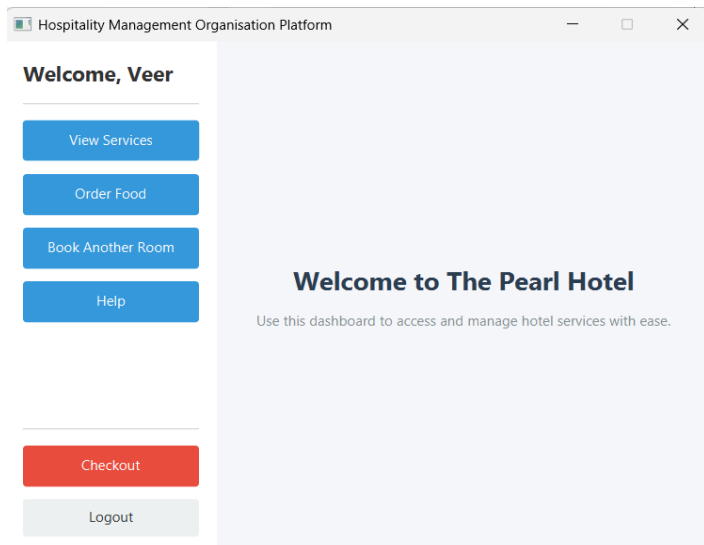
Reservation Successful

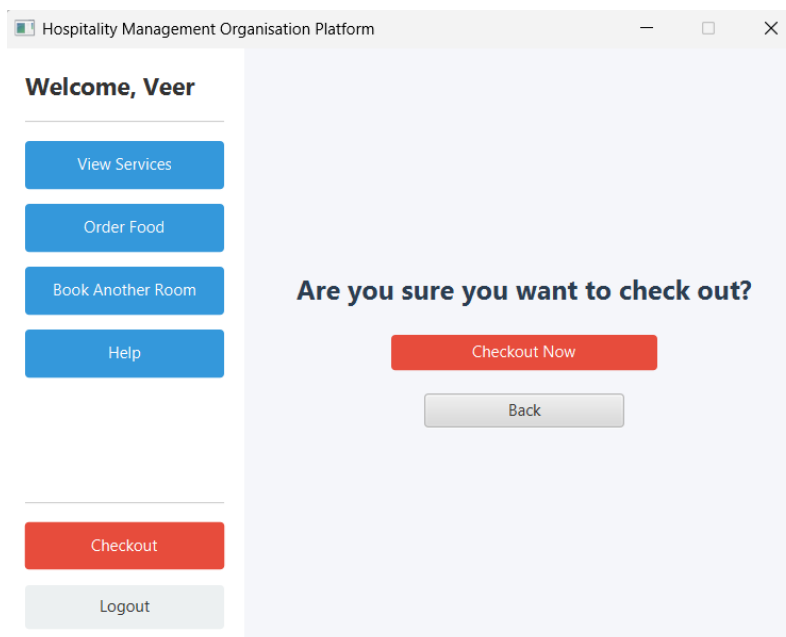
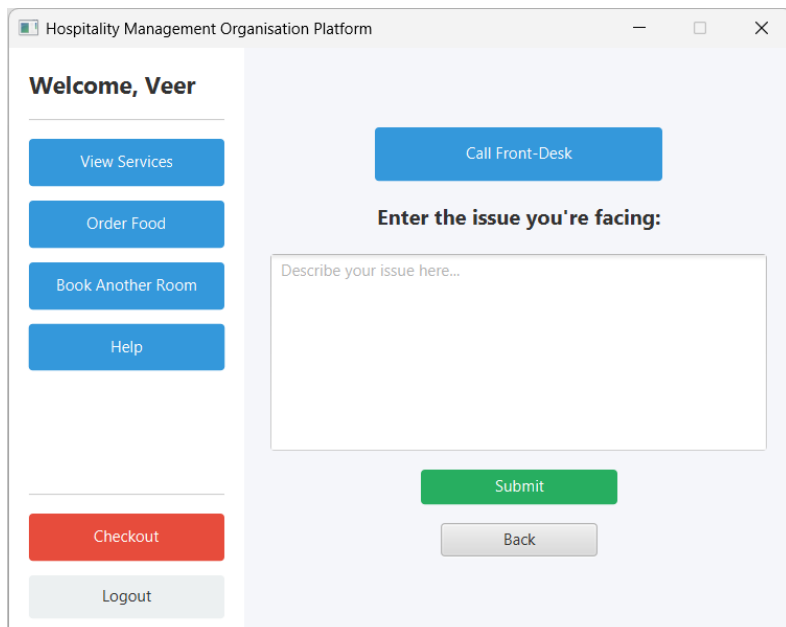
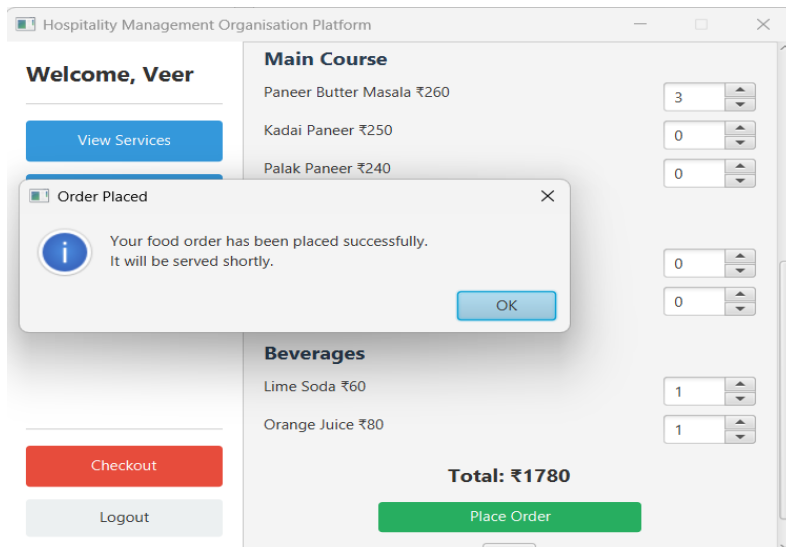
i

Your booking has been confirmed!

Booking ID: 52220
Room Numbers: 101, 102, 103, 104, 201, 202, 301, 302

OK





Hospitality Management Organisation Platform

Select Your Room

Customize your stay with options below

Single Bed • ₹800/Night

14 available

Category

☐ Standard

☒ Premium (+₹600)

☐ Deluxe (+₹1200)

☐ AC (+₹500)

Qty

Double Bed • ₹1200/Night

31 available

Category

☒ Standard

☐ Premium (+₹600)

☐ Deluxe (+₹1200)

☐ AC (+₹500)

Qty

Triple Bed • ₹1800/Night

26 available

Category

☒ Standard

☐ Premium (+₹600)

☐ Deluxe (+₹1200)

☐ AC (+₹500)

Qty

Stay Duration

From To

Hospitality Management Organisation...

Guest Details

Enter your information

Full Name

Date of Birth

Phone Number

Error

☒ You must be at least 18 years old to book.

Confirm Password

Hospitality Management Organisation...

Guest Details

Enter your information

Full Name

Date of Birth

Error

☒ Passwords do not match.

Set Password

Confirm Password

Hospitality Management Organisati...

Guest Details

Enter your information

Full Name
Gaurav

Date of Birth
6/19/2001

Phone Number
9856423

Address
#20 Nira

Set Password
...

Confirm Password
...

Back Book

Error

Phone number must be exactly 10 digits.

OK

Database Monitoring and Testing

The project includes SQL validation queries for system monitoring.

1. Check all bookings

hotel_db/postgres@PostgreSQL 18

Query Query History

```
1 SELECT * FROM bookings;
```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	booking_id [PK] character varying (20)	name character varying (100)	phone character varying (20)	password character varying (100)	dob date	address text	from_date date	to_date date	status character varying
1	58636	Raj	6985532451	111	1988-06-...	45 Green Park Avenue, Mumbai, Maharashtra 4...	2026-06-11	2026-06-...	ACTIVE
2	97776	Kumar	9865423555	111	1999-06-...	78 MG Road Extension, Pune, Maharashtra 411...	2026-06-18	2026-06-...	ACTIVE
3	31561	Vikas	8645175362	111	1986-06-...	156 Palm Street, Chennai, Tamil Nadu 600002	2026-06-18	2026-06-...	ACTIVE
4	56868	Komal	9546217563	111	2002-06-...	9 Silver Heights, Jaipur, Rajasthan 302001	2026-06-17	2026-06-...	ACTIVE
5	16943	Smith	8965752365	111	1992-06-...	34 Sunrise Colony, Lucknow, Uttar Pradesh 226...	2026-06-06	2026-06-...	ACTIVE
6	82572	Anaya	9685432141	111	2000-06-...	12 Lake View Residency, Bengaluru, Karnataka ...	2026-06-06	2026-06-...	CHECKED_OUT
7	93853	Srinivas	8965427325	111	2001-06-...	23 Rosewood Apartments, Hyderabad, Telanga...	2026-06-12	2026-06-...	CHECKED_OUT

2. Check a specific booking

hotel_db/postgres@PostgreSQL 18

Query Query History

```
1 SELECT *
2 FROM bookings
3 WHERE booking_id='93853';
4
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	booking_id [PK] character varying (20)	name character varying (100)	phone character varying (20)	password character varying (100)	dob date	address text	from_date date	to_date date	status character
1	93853	Srinivas	8965427325	111	2001-06-...	23 Rosewood Apartments, Hyderabad, Telangana 500...	2026-06-12	2026-06-...	ACTIVE

3. Check all room assignments

hotel_db/postgres@PostgreSQL 18

Query Query History

```
1 SELECT * FROM booked_rooms;
```

Data Output Messages Notifications

	id [PK] integer	booking_id character varying (20)	room_no integer
1	179	58636	101
2	180	58636	102
3	181	58636	103
4	182	58636	104
5	183	58636	105
6	184	58636	201
7	185	58636	202
8	186	58636	301
9	187	58636	302
10	188	82572	106
11	189	82572	107
12	190	82572	108
13	191	82572	109
14	192	82572	110

Total rows: 67 Query complete 00:00:00.142

4. Check rooms allotted for a booking

hotel_db/postgres@PostgreSQL 18

Query Query History

```
1 SELECT *  
2 FROM booked_rooms  
3 WHERE booking_id='58636';  
4
```

Data Output Messages Notifications

	id [PK] integer	booking_id character varying (20)	room_no integer
1	179	58636	101
2	180	58636	102
3	181	58636	103
4	182	58636	104
5	183	58636	105
6	184	58636	201
7	185	58636	202
8	186	58636	301
9	187	58636	302

5. Check all Available rooms

hotel_db/postgres@PostgreSQL 18

Query Query History

```
1 SELECT *  
2 FROM rooms  
3 ORDER BY room_no;  
4
```

Data Output Messages Notifications

	room_no [PK] integer	room_type character varying (20)	is_available boolean
21	121	SINGLE	false
22	122	SINGLE	false
23	123	SINGLE	false
24	124	SINGLE	false
25	125	SINGLE	false
26	126	SINGLE	true
27	127	SINGLE	true
28	128	SINGLE	true
29	129	SINGLE	true
30	130	SINGLE	true
31	201	DOUBLE	false
32	202	DOUBLE	false
33	203	DOUBLE	false
34	204	DOUBLE	false

6. Check total available room types

hotel_db/postgres@PostgreSQL 18

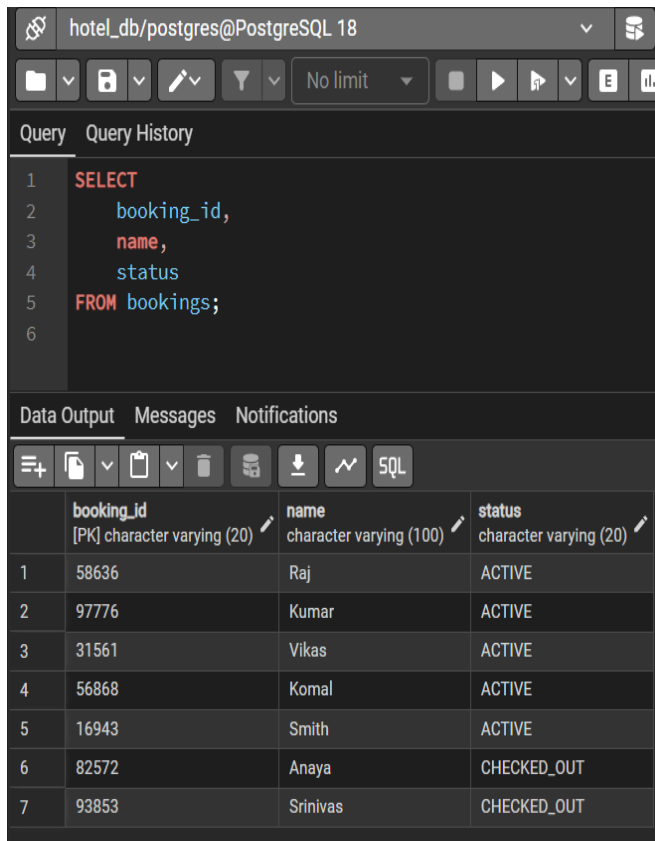
Query Query History

```
1 SELECT room_type, COUNT(*)  
2 FROM rooms  
3 GROUP BY room_type;  
4
```

Data Output Messages Notifications

	room_type character varying (20)	count bigint
1	TRIPLE	30
2	DOUBLE	40
3	SINGLE	30

7. Check booking status



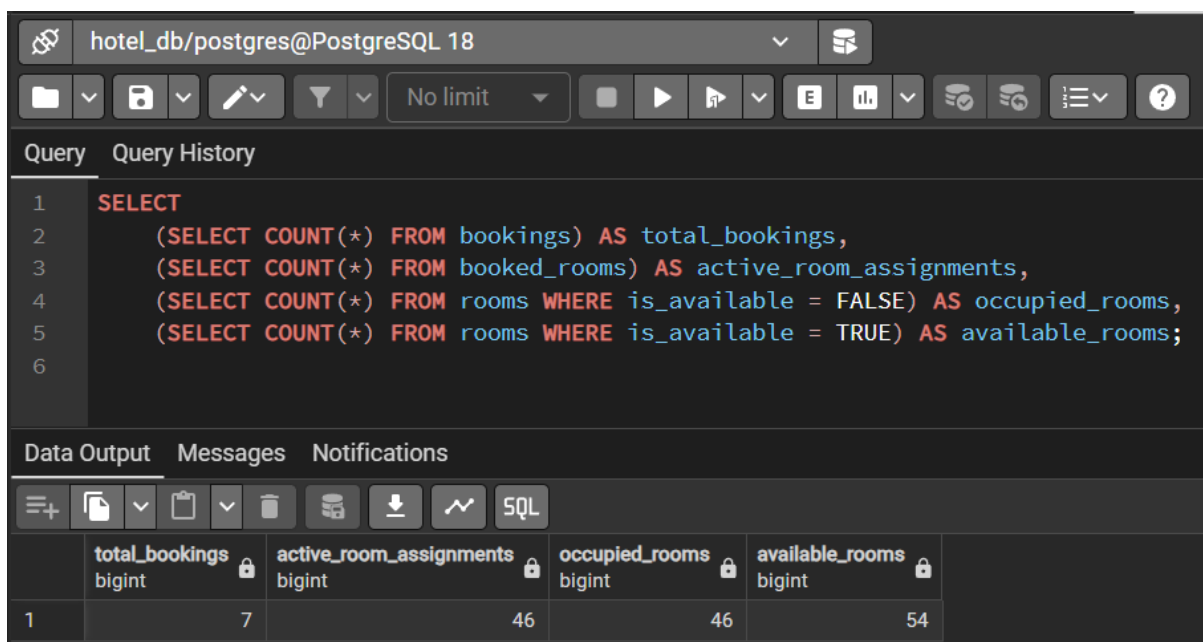
The screenshot shows a PostgreSQL client window titled 'hotel_db/postgres@PostgreSQL 18'. The query editor contains the following SQL query:

```
1 SELECT
2     booking_id,
3     name,
4     status
5 FROM bookings;
6
```

The 'Data Output' tab is selected, displaying the results of the query in a table. The table has three columns: 'booking_id' (PK), 'name', and 'status'. The results are as follows:

	booking_id [PK] character varying (20)	name character varying (100)	status character varying (20)
1	58636	Raj	ACTIVE
2	97776	Kumar	ACTIVE
3	31561	Vikas	ACTIVE
4	56868	Komal	ACTIVE
5	16943	Smith	ACTIVE
6	82572	Anaya	CHECKED_OUT
7	93853	Srinivas	CHECKED_OUT

8. Quick project health check



The screenshot shows a PostgreSQL client window titled 'hotel_db/postgres@PostgreSQL 18'. The query editor contains the following SQL query:

```
1 SELECT
2     (SELECT COUNT(*) FROM bookings) AS total_bookings,
3     (SELECT COUNT(*) FROM booked_rooms) AS active_room_assignments,
4     (SELECT COUNT(*) FROM rooms WHERE is_available = FALSE) AS occupied_rooms,
5     (SELECT COUNT(*) FROM rooms WHERE is_available = TRUE) AS available_rooms;
6
```

The 'Data Output' tab is selected, displaying the results of the query in a table. The table has four columns: 'total_bookings', 'active_room_assignments', 'occupied_rooms', and 'available_rooms', all of type 'bigint'. The results are as follows:

	total_bookings bigint	active_room_assignments bigint	occupied_rooms bigint	available_rooms bigint
1	7	46	46	54

Future Enhancements

- Online Payment Gateway Integration for secure room booking payments.
- Email and SMS notifications for booking confirmations, reminders, and checkout updates.
- Real-Time Analytics and Reporting for occupancy rates, revenue tracking, and booking trends.
- Cloud Database Deployment to enable access from multiple devices and locations.
- Customer Feedback and Rating System to improve service quality.
- Dynamic Pricing System based on room demand, season, and occupancy levels.
- Integration with External Booking Platforms such as Booking.com and MakeMyTrip.

Conclusion

The Hotel Management Platform successfully automates hotel reservation and room management operations through an intuitive JavaFX-based user interface and a PostgreSQL backend database. The system enables customers to register, authenticate, book rooms, view reservation details, access hotel services, and manage bookings efficiently.

The project demonstrates the practical application of Object-Oriented Programming principles, JDBC database connectivity, JavaFX GUI development, and PostgreSQL database management. By integrating frontend functionality with persistent backend storage, the platform provides a reliable and scalable solution for handling hotel operations.

Overall, the project serves as a comprehensive example of desktop application development and highlights the effective use of modern software engineering concepts in solving real-world hotel management challenges.